USN | | | | | | | | | | [22CE32]

THIRD SEMESTER B. Tech EXAMINATION, DECEMBER - 2023

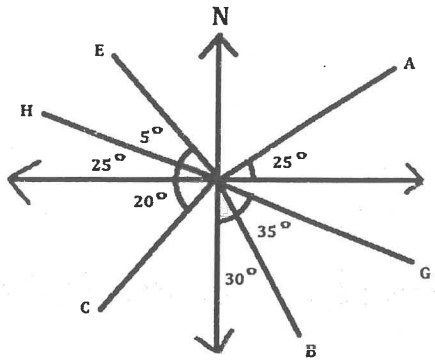
SURVEYING - I

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART - A			**5 X 2=10**
1.	**a.**	Explain geodetic surveying.	L2	CO1,2	**02**
	b.	What are the three main methods used in plane table surveying?	L2	CO1,2	**02**
	c.	Name three instruments used to find levels in surveying.	L2	CO1,2	**02**
	d.	How can you calculate the area of an irregular figure using the average ordinate rule?	L2	CO1,2	**02**
	e.	Define contour interval in the context of contouring.	L2	CO1,2	**02**
		PART - B			**6 X 5=30**
2.	**a.**	Explain chain surveying.	L2	CO1,2	**05**
		OR			
	b.	Give a brief note on the different Instruments Required For Linear Measurement.	L3	CO2,3	**05**
3.	**a.**	List the different temporary adjustments of Plane Table Surveying.	L2	CO2	**05**
		OR			
	b.	Explain with a neat sketch the whole circle and quadrant circle bearing systems.	L2	CO2	**05**
4.	**a.**	The following consecutive readings were taken with a level and 5M leveling staff on continuously sloping ground at a common interval of 20M. 0.385, 1.030, 1.925, 2.825, 3.730, 4.685, 0.625, 2.005, 3.110, 4.485. The Reduced Level of the first point was 208.125M. Rule out a page of a level field book and enter the above readings. Calculate the reduced levels of the points by Rise and Fall Method and also the Gradient of the line joining the first and the last point.	L3	CO3,4	**05**
		OR			
	b.	List and explain the different types of leveling.	L2	CO2	**05**

5.	**a.**	The following perpendicular offsets were taken at 10 meters' intervals from a survey line to an irregular boundary line. 3.25, 5.60, 4.20, 6.65, 8.75, 6.20, 3.25, 4.20, 5.65 Calculate the areas enclosed between all the lines using Average Ordinate Rule.	L3	CO3,4	**05**
		OR			
	b.	The following perpendicular offsets were taken at 10 meters' intervals from a survey line to an irregular boundary line. 3.25, 5.60, 4.20, 6.65, 8.75, 6.20, 3.25, 4.20, 5.65 Calculate the areas enclosed between all the lines using Trapezoidal Rule.	L3	CO3,4	**05**
6.	**a.**	List the different characteristics of contour cines.	L2	CO2	**05**
		OR			
	b.	Give a brief note on methods of interpolation in contouring.	L4	CO5	**05**
7.	**a.**	Enumerate and elucidate minor tools for establishing right angles.	L3	CO2,3	**05**
		OR			
	b.	Differentiate between the different Types of compass used in surveying.	L3	CO2,3	**05**
		PART - C			**6 X 10=60**
8.	**a.**	Explain the principles of surveying.	L4	CO5	**10**
		OR			
	b.	List and explain the different classification of surveying based on instruments	L4	CO6	**10**
9.	**a.**	Convert the angles in to whole circle bearing and quadrant bearing for the given figure  	L4	CO5	**10**
		OR			
	b.	Devise a comprehensive procedure for acquiring ground data utilizing Intersection methods, including implementing temporary adjustments	L4	CO4,6	**10**

10.	a.	A page of a level book was defaced so that the only legible figures were Consecutive entries in the column of reduced level, 55.565(B.M), 54.985(TP), 55.170, 56.265, 53.670, 53.940(TP), 52.180, 52.015, 51.480, (TP), 53.145, 54.065(TBM). Entries in Back Sight Column, 1.545, 2.310, 0.105, 3.360 in order from top of the page. Reconstruct the page as booked and check your work.	L4	CO5	**10**
		OR			
	b.	The following figures were extracted from a level field book. Some of the entries being illegible owing to exposure to rain. Insert the missing figures and check your results. Re-book all the figures by the Rise and Fall Method.	L4	CO5,6	**10**

Station	BS	IS	FS	RISE	FALL	R. L	Remark
A	2.285					232.460	BM 1
B	1.650		X	0.020			
		2.105			X		
C	X		1.960	X			
D	2.050		1.925		0.300		
		X		X		232.255	BM 2
E	1.690		X	0.340			
F	2.865		2.100		X		
			X	X		233.425	BM 3

11.	a.	The following perpendicular offsets were taken from a chain line to a hedge	L4	CO5	**10**

CHAINAGE (M)	OFFSETS (M)
0	7.6
15	8.5
30	10.7
45	12.8
60	10.6
70	9.5
80	8.3
100	7.9
120	6.4
140	4.4

Calculate the area between all the lines by using Simpson's Rule.

OR

	b.	The Following perpendicular offsets were taken from a chain line to an irregular boundary.	L4	CO5	**10**

CHAINAGE (M)	0	10	25	42	60	75
OFFSET (M)	15.5	26.2	31.8	25.6	29.0	31.5

		Calculate the area between the chain line, the boundary and the end offsets by trapezoidal rule.			
12.	a.	Give a brief note on L-section in contouring.	L4	CO5	**10**
		OR			
	b.	Explain the indirect methods of contouring.	L4	CO5	**10**
13.	a.	The following perpendicular offsets were taken from a chain line to a hedge CHAINAGE (M): 0, 15, 30, 45, 60, 70, 80, 100, 120, 140. OFFSETS (M): 7.6, 8.5, 10.7, 12.8, 10.6, 9.5, 8.3, 7.9, 6.4 and 4.4. Calculate the area between all the lines by using Trapezoidal Rule.	L4	CO5	**10**
		OR			
	b.	List and explain the different methods of running a theodolite traverses.	L4	CO4,6	**10**